UNIT - V

Design of Bearings and Gears

1. Sliding Contact (Journal) Bearings

A bearing supports a rotating / oscillating shaft and carries load while allowing relative motion with low friction. Sliding (plain) bearing : journal rubs in a sleeve with a lubricant film in between.

- Types : journal (radial load) - full 360°, partial (120°-180°), fitted ; thrust (axial) - collar, pivoted pad (Michell / Kingsbury) ; hydrodynamic (film from motion) , hydrostatic (film from external pump).
- Merits vs rolling : silent, withstands shock & high load, compact radially, cheap for large sizes, long life with film, split for crank-shaft ; Demerits : high starting friction, needs continuous oil supply, axial space ↑.

Modes (regimes) of lubrication

Mode	Film	μ	Remarks
Hydrodynamic (thick film)	full film, surfaces separated	0.001 - 0.01	needs speed, $\mu \propto Z N / p$
Mixed / thin film	partial asperity contact	0.01 - 0.1	start-up / low speed
Boundary	molecular oil layer only	0.05 - 0.2	oiliness, additives
Hydrostatic	pressurised oil from pump	≈ 0.001	works at zero speed
Elasto-hydrodynamic	film in highly loaded contact	0.001 - 0.005	gears, rolling bearings

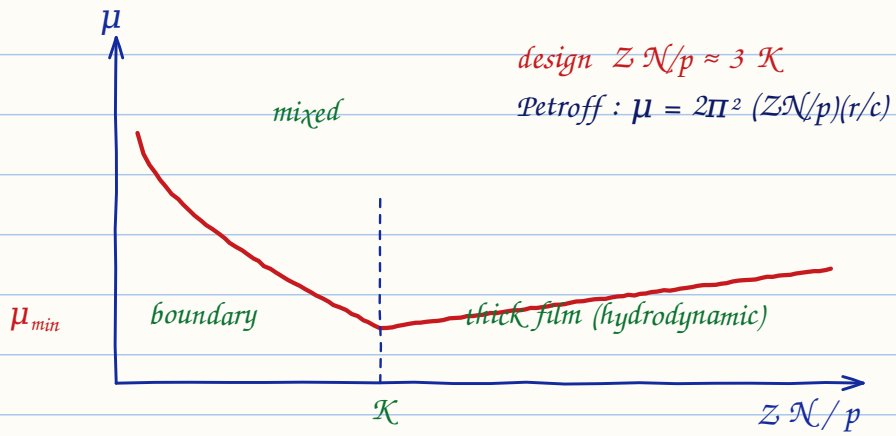


Fig 1: Stribeck curve - friction vs bearing characteristic $Z N / p$

- Bearing characteristic number ZN/p (Z = absolute viscosity Pa·s, N rpm, $p = W/(l\,d)$ bearing pressure). Bearing modulus K = value of ZN/p at minimum friction ; to the left of K the film breaks (mixed) → design with $ZN/p \geq 3K$ as safety, usually ZN/p well in the thick-film zone.
- Sommerfeld number $S = (ZN/p)(r/c)^2$ (dimensionless, N in rps, p in Pa) - basic design variable ; Raimondi - Boyd charts relate S to ε, h_0/c, μ r/c, flow.

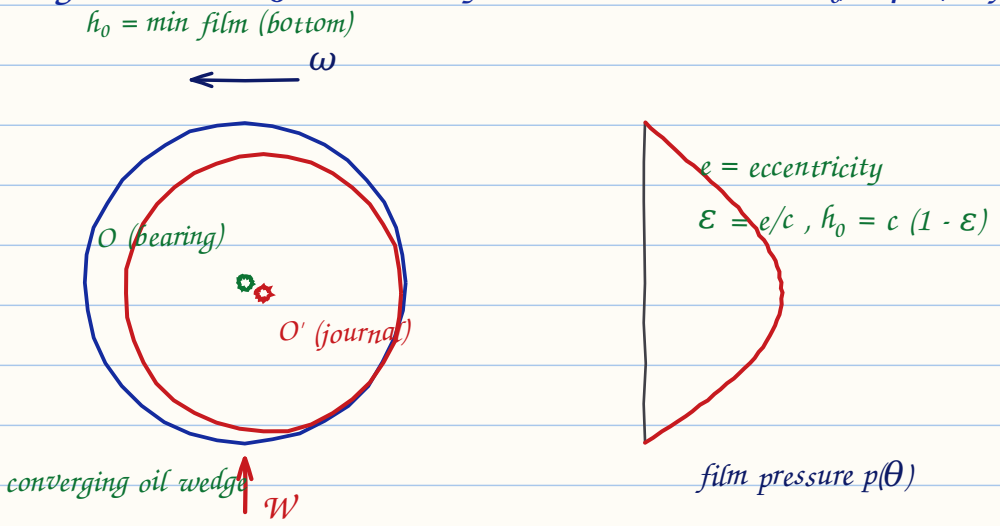


Fig 2: Hydrodynamic journal bearing - oil wedge

At rest the journal touches the bearing at the bottom. On rotation, oil is dragged into the converging wedge and a pressure builds in it that lifts the journal and shifts it sideways (attitude angle) : the shaft floats on oil. Minimum film thickness $h_0 = c\,(1 - \varepsilon)$ must exceed the surface roughness ($h_0 \geq 0.005$ mm).

Terminology

- Journal radius r , bearing radius R ; radial clearance $c = R - r$ ($c/r = 0.001$ to 0.002 or $c = 0.001\,d$) ; eccentricity e ; eccentricity ratio $\varepsilon = e/c$; l/d ratio = 0.75-2.8 (usually 1-1.5 : long → end leakage ↓, misalignment ↑) ; projected area $A = l\,d$.

Application	p (N/mm²)	l/d	Application	p (N/mm²)
Electric motor	0.7 - 1.4	1 - 2	Railway axle	3.5 - 4.5
Gear box / turbine	0.7 - 2	1 - 2	Crank pin (engine)	7 - 14
Machine tool	0.7 - 1.4	1 - 4	Main bearing (auto)	4 - 5

2. Friction in Journal Bearings - Petroff & McKee

Petroff's equation assumes the shaft is concentric (no eccentricity, no load) so the friction is just the viscous shear of the oil film :

$$\mu = 2\pi^2 (Z N / p)(r / c) \quad (N \text{ in rps , SI units})$$

$$\text{McKee : } \mu = (33 / 10^8)(Z N / p)(r / c) + k$$

- In McKee's form Z in kg/m·s (Pa·s), N in rpm, p in N/mm² ; k = correction for end leakage ≈ 0.002 for l/d = 0.75 to 2.8. The two give the straight line of the Stribeck curve ; experiments (McKee brothers) confirmed for ZN/p beyond the transition.
- Power lost in friction : $H_g = \mu W v$ (W) , $v = \pi d N / 60$ (m/s) ; torque $T_f = \mu W r$.

Heat balance (thermal equilibrium)

$$H_g = \mu W v \text{ (generated)} \le H_d = C A (t_b - t_a)$$

$$A = l d \text{ (projected area)}$$

C = heat dissipation coefficient : 140-420 W/m²·°C (unventilated), up to 1230 (well ventilated).

- Bearing temperature $t_b = t_a + (H_g / C A) \le$ 60-70°C ; oil film temperature ≈ $(t_b + t_o)/2$ where viscosity Z is taken. If $H_g > H_d \rightarrow$ increase area, use oil circulation / cooler, reduce μ.

Design procedure for a hydrodynamic journal bearing

- (1) Given W, N, d (from shaft design). (2) Choose $l/d \rightarrow l$; $p = W/(l\,d) \le p_{allow}$. (3) Select lubricant ($Z$ at operating temperature, SAE grade). (4) c/r (or c = 0.001 d). (5) Check $ZN/p \ge 3K$. (6) μ by McKee. (7) $H_g = \mu W v$; (8) $H_d = C A \Delta t \rightarrow t_b$; keep ≤ 70°C ; (9) check h_0 and ε from charts.

Example 1 : d = 50 mm, l = 75 mm (l/d = 1.5), W = 3 kN, N = 600 rpm, Z = 0.025 Pa·s, c/r = 0.001 (r/c = 1000), ambient 30°C, C = 300 W/m²°C. $p = 3000/(50 \times 75) = 0.8$ N/mm². $ZN/p = 0.025 \times 600/0.8 = 18.75$. $\mu = 33\times10^{-8} \times 18.75 \times 1000 + 0.002 =$ 0.0082. $v = \pi \times 0.05 \times 600/60 = 1.57$ m/s ; $H_g = 0.0082 \times 3000 \times 1.57 = 38.6$ W. $H_d/\Delta t = 300 \times (0.05 \times 0.075) = 1.125$ W/°C $\rightarrow \Delta t = 34$°C $\rightarrow t_b = 64$°C ≤ 70°C (OK).

Lubricants

- Absolute viscosity Z (Pa·s ; 1 cP = 10^{-3} Pa·s) ; kinematic $\nu = Z/\rho$ (cSt ; $\rho \approx 870$ kg/m³) ; viscosity falls sharply with temperature - measured by viscosity index VI (high VI = small change). SAE grades 10, 20, 30, 40, 50 (higher = thicker, for slower speed / heavier load / hotter).
- Other properties : oiliness (boundary film), flash & fire point (safety), pour point (cold start), oxidation stability, anti-wear / EP additives. Grease (oil + soap) for low speed, intermittent, difficult to feed ; solid lubricants (graphite, MoS_2) for very high temperature / vacuum.
- Supply : ring oiling, splash, bath, pressure feed through groove at the unloaded side (oil must enter where film is thick, not in the loaded zone) ; axial grooves must not run in the loaded region.

Hydrostatic & thrust bearings (short note)

- Hydrostatic : oil at pressure p_s from a pump fills pockets in the bearing → $W = p_s A_{eff}$; film exists at zero speed → no wear at start-up, very low friction, high stiffness, but pump + filter + piping cost. Used in machine tool spindles, telescopes, heavy turbines (jacking oil).
- Thrust bearing : flat collar cannot form wedge → pivoted tilting pad (Michell / Kingsbury) : pad tilts to make a wedge film ; capacity $W = p A$; ship propeller, turbines, vertical pumps. Step / tapered land bearings for small sizes.

3. Bearing Materials & Failures

- Requirements : compressive & fatigue strength, conformability (take misalignment), embeddability (bury dirt), low friction, thermal conductivity, corrosion & seizure resistance, cheap.

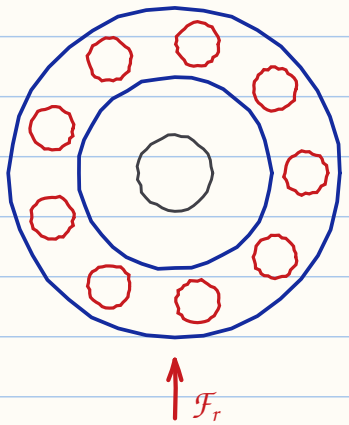
Material	Properties	Use
Tin babbitt (Sn 90, Sb 7, Cu 3)	soft, best embeddability / conformability, weak > 120°C	turbines, general
Lead babbitt	cheaper, lower strength	light duty
Copper-lead / lead bronze	high fatigue strength, to 250°C	IC engines, heavy
Phosphor / tin bronze	strong, hard shaft needed	gears, worm shafts
Cast iron	low cost, slow speeds, good with grease	machine tools
Sintered (oil-impregnated)	self-lubricating, no maintenance	fans, appliances
PTFE / nylon / carbon	dry running, corrosion proof	chemical, food

Bearing failures

- Abrasion / scoring (dirt, rough shaft) → filters, seals, hard smooth journal ($Ra \leq 0.4\ \mu m$) ; fatigue (cracks, pitting) (cyclic pressure) → thin lining ; wiping / seizure (film breakdown, overheating) → proper cooling, correct clearance ; corrosion (acid in oil) → inhibitors ; cavitation / erosion ; edge loading (misalignment) → spherical seat, relief ; fretting.

4. Rolling Contact Bearings

Load transmitted through balls / rollers between two hardened (≥ 60 HRC, 100Cr6 / SAE 52100) rings, rolling friction ≈ 0.001-0.005 : low starting friction, accurate position, standardised (ISO), interchangeable, but noisy, poor with shock, limited life (fatigue), larger radius.



heaviest ball $W_{max} \approx 5\ F_r/z$

Stribeck: $C_0 = k\, d^2\, z\, /\, 5$

$L_{10} = (C/P)^3$ million rev (balls)

$= (C/P)^{10/3}$ (rollers)

$P = X\, V\, F_r + Y\, F_a$

Fig 3: Deep groove ball bearing & load relations

Type	Load	Remarks
Deep groove ball (6xxx)	radial + some axial both ways	most common, high speed
Angular contact (7xxx)	radial + large axial (one way)	paired / preloaded, spindles
Self-aligning ball (1xxx)	radial, tolerate 2-3° misalignment	line shafts
Cylindrical roller (NU, N)	heavy radial, no axial (NJ small)	motors, gearboxes
Tapered roller (3xxx)	heavy radial + axial one way	wheel hubs, pairs
Spherical roller (2xxx)	very heavy + misalignment	mining, rolls
Thrust ball / roller (5xxx)	axial only	pivots, vertical shafts

Bearing designation (ISO)

e.g. 6205 - 2RS : 6 = deep groove ball ; 2 = light series (dimension / width) ; 05 → bore = 05 × 5 = 25 mm (04 → 20 mm ; for bore 20-480 mm : last two digits × 5 ; 00 → 10 , 01 → 12 , 02 → 15 , 03 → 17 mm). Suffix : Z shield , 2RS rubber seals , C3 larger clearance.

- Fits : rotating ring (inner for rotating shaft) - interference fit (k5 / m5) on shaft ; stationary ring - transition / slip fit in housing (H7) to allow creep ; shoulders ≥ fillet radius of bearing ; axial location with circlips / nut.
- Failures : subsurface fatigue spalling (normal life end), brinelling (static / shock overload), false brinelling, wear (dirt), corrosion, cage break, smearing (skid), electrical fluting. Lubrication : grease (≤ 0.3-0.5 × limiting speed, sealed-for-life) or oil bath / mist (high speed).

5. Static & Dynamic Load Capacity

Static load capacity C_0 - Stribeck's equation

Static capacity = radial load for which the permanent deformation at the most heavily loaded rolling element contact = 0.0001 × diameter of rolling element (does not impair smooth running). Stribeck : the max load on a ball $W_{max} = 5\ W/z$ (clearance) ≈ $4.37\ W/z$ (no clearance) ; W_{max} limited by Hertz contact stress → $W_{max} = k\ d^2$ →

$$C_0 = k\ d^2\ z\ /\ 5 \quad \text{(ball)} \quad ; \quad C_0 = k\ l\ d\ z\ /\ 5 \quad \text{(cylindrical roller)}$$

static safety : $C_0 \geq s_0\ P_0$ (s_0 = 1 normal , 1.5-2 for shock / quiet)

k = constant depending on materials and radii of curvature of contact (N/mm^2) ; d = ball dia ; z = number of balls ; l = roller length. Equivalent static load $P_0 = 0.6\ F_r + 0.5\ F_a$ ($\geq F_r$) for deep groove ball bearing.

Dynamic load capacity C & L_{10} life

- Basic dynamic capacity C = constant radial load for which 90% of a group of identical bearings complete 10^6 revolutions without fatigue failure (rating life L_{10}). Catalogue gives C and C_0.

$$L_{10} = (C / P)^k \text{ million rev ; } k = 3 \text{ (ball), } 10/3 \text{ (roller)}$$
$$L_{10h} = 10^6\ (C/P)^k / 60\ N \text{ (hours)}$$
$$C_{required} = P\ (L_{10})^{1/k}\ ,\ L_{10} = 60\ N\ L_{10h}/10^6$$

- Load-life : doubling P cuts the ball bearing life to 1/8 (cubic law). Reliability other than 90% : $L_R = a_1\ L_{10}$; a_1 = 0.62 (95%), 0.53 (96%), 0.44 (97%), 0.33 (98%), 0.21 (99%).
- Variable loads / speeds : $P_{eq} = (\Sigma\ P_i^3\ n_i / \Sigma\ n_i)^{1/3}$; equivalent speed $N_{eq} = \Sigma\ N_i\ t_i$; Palmgren-Miner : $\Sigma\ (n_i/L_i) = 1/L$.

Machine	L_{10h} (h)
Household / agricultural / instruments	300 - 3000
Light electric motors, fans (8 h/day)	8000 - 15000
Gear boxes, machine tools (8 h/day)	20000 - 30000
Pumps, compressors (24 h/day)	40000 - 60000
Power-plant, mines - continuous	100000+

Equivalent bearing load

$$P = K_a\ (X\ V\ F_r + Y\ F_a)$$

- V = 1 (inner ring rotates), 1.2 (outer ring rotates) ; load (service) factor K_a : steady 1.0-1.1 ; light shock 1.2-1.5 ; moderate shock 1.5-2 ; heavy shock 2-3. X, Y from catalogue : if $F_a/(V\ F_r) \leq e \rightarrow X = 1$, $Y = 0$; if $> e \rightarrow X = 0.56$ and Y from table (deep groove) :

F_a/C_0	0.025	0.04	0.07	0.13	0.25	0.5
e	0.22	0.24	0.27	0.31	0.37	0.44
Y	2.0	1.8	1.6	1.4	1.2	1.0

Example 2a (variable load) : a ball bearing (C = 25 kN) runs at 800 rpm : 5 kN for 40% of time, 3 kN for 35%, 1.5 kN for 25%. $P_{eq} = (5^3 \times 0.4 + 3^3 \times 0.35 + 1.5^3 \times 0.25)^{1/3} = (60.3)^{1/3} = 3.92$ *kN.* $L_{10} = (25/3.92)^3 = 259$ *million rev →* $L_{10h} = 259\times10^6/(60 \times 800) = 5400$ *h.*

6. Selection of Bearings from Catalogue

- *(1) Radial* F_r*, axial* F_a*, speed* N*, shaft dia d (bore), required* L_{10h}*, reliability, environment. (2) Choose type (ball for light / high speed, roller for heavy, tapered for combined, thrust for pure axial, self-aligning if misaligned). (3)* $L_{10} = 60\,N\,L_{10h}/10^6$*. (4)* $P = K_a(X\,V\,F_r + Y\,F_a)$ *(guess* C_0 *first, iterate). (5)* $C_{req} = P\,L_{10}^{1/k}$*. (6) Select bearing with bore d and* $C \geq C_{req}$*. (7) Check* $C_0 \geq s_0\,P_0$ *and limiting speed.*

Example 2 : shaft d = 40 mm, $F_r = 4$ *kN,* $F_a = 1.5$ *kN,* $N = 1000$ *rpm,* $L_{10h} = 5000$ *h, inner ring rotates, moderate shock* $K_a = 1.2$*.* $L_{10} = 60 \times 1000 \times 5000/10^6 = 300$ *million rev. Try 6308 (typical C = 41 kN,* $C_0 = 24$ *kN) :* $F_a/C_0 = 0.0625 \rightarrow e \approx 0.26$ *;* $F_a/F_r = 0.375 > e \rightarrow X = 0.56$*,* $Y \approx 1.7$*.* $P = 1.2\,(0.56 \times 4 + 1.7 \times 1.5) = 5.75$ *kN.* $C_{req} = 5.75 \times 300^{1/3} = 5.75 \times 6.69 = 38.5$ *kN ≤ 41 (OK) → 6308 is safe (check against actual catalogue values).*

7. Spur Gears - Terminology & Tooth Forces

Spur gear : straight teeth parallel to axis, transmits power between parallel shafts. Preferred module series (mm) : 1 , 1.25 , 1.5 , 2 , 2.5 , 3 , 4 , 5 , 6 , 8 , 10 , 12 , 16 , 20 .

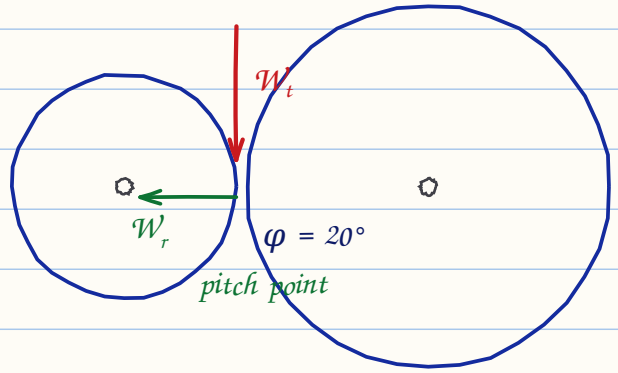
$$\text{module } m = d/z \quad ; \quad \text{circular pitch } p_c = \pi m \quad ; \quad d = m z$$

$$\text{addendum} = m \; ; \; \text{dedendum} = 1.25\,m \; ; \; \text{clearance } 0.25\,m \; ; \; \text{tooth depth } 2.25\,m$$

$$i = N_1/N_2 = z_2/z_1 \quad ; \quad \text{centre distance } a = m\,(z_1 + z_2)/2$$

- *Pressure angle* $\varphi = 20°$ *(full-depth involute) , 14.5° , 20° stub ; minimum teeth to avoid interference* $z_{min} = 2/\sin^2\varphi = 17\ (20°),\ 32\ (14.5°)$ *; face width b = 9.5-12.5 m (usually 10 m) ;* $z_{pinion} \geq 18$*-20 for steel.*

- Materials : cast iron FG 200-260 (σ_b ≈ 50-70 MPa) ; C45 / 40C8 (σ_b 140-200) ; alloy steel 40Ni14, 35Mn2Mo28 ; case hardened 20MnCr5 (BHN 600) ; cast steel CS 540 ; phosphor bronze (worm wheel) ; nylon / fibre (silent).



$W_t = 2T/d = P/v$

$W_r = W_t \tan \varphi$

$W_n = W_t / \cos \varphi$

$v = \pi d N / 60$

Fig 4: Forces on a spur gear tooth at the pitch point

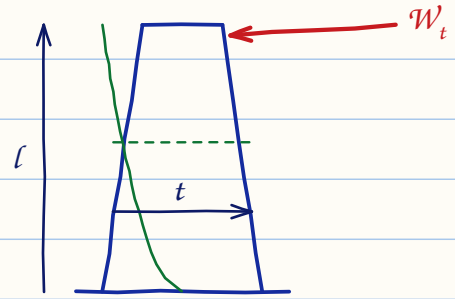
- Failure modes : tooth breakage (bending fatigue at root - beam strength), pitting (surface contact fatigue - wear strength), scoring (scuffing - lubrication), abrasive wear, plastic flow (soft gears).

Involute tooth profile - key facts

- Involute = curve traced by a point on a taut string unwound from the base circle ($d_b = d \cos \varphi$). Conjugate action : constant velocity ratio even if centre distance changes slightly ; line of action tangent to both base circles passes through the pitch point at angle φ ; teeth contact along the line of action only.
- Contact ratio = length of path of contact / base pitch ≥ 1.2 (at least 1.4 desirable) - average number of teeth in mesh ; 20° gears ≈ 1.6-1.8. Smooth, quiet, less load per tooth if > 1.
- Interference : tip of gear tooth cuts the root of pinion below the base circle → weak tooth. Avoided by $z \geq z_{min}$ (17 for 20°), stub teeth, increasing φ, or profile shift (corrected gears). Undercutting in hobbing small pinions weakens the root.
- Cycloidal profile : no interference but needs exact centre distance, costly - used in watches. Gear quality (error e) by class : precision ground (e ≈ 0.01 mm) → commercial cut (0.1 mm).

8. Beam Strength - Lewis Equation

Lewis (1892) : tooth is a cantilever beam of uniform strength (a parabola inscribed in the tooth profile) loaded by the tangential component W_t at the tip (assumes full load on one tooth, no impact).



tooth = cantilever, critical section t at root

$\sigma_b = 6\, W_t\, l / (b\, t^2)$

$W_t = \sigma_b\, b\, m\, Y$ (Lewis)

$Y = \pi\, y = 0.484 - 2.87/z$ (20° full depth)

Fig 5: Lewis model of a gear tooth

$$\sigma_b = 6\, W_t\, l / b\, t^2 = W_t / (b\, m\, Y)$$

Beam strength $S_b = \sigma_b\, b\, m\, Y$ (N) ≥ effective load F_{eff}

Tooth system	Form factor y (Lewis)	$Y = \pi\, y$
20° full depth	$y = 0.154 - 0.912/z$	$0.484 - 2.87/z$
14.5° composite / full depth	$y = 0.124 - 0.684/z$	$0.390 - 2.15/z$
20° stub	$y = 0.175 - 0.841/z$	$0.55 - 2.64/z$

- Pinion vs gear : the weaker of the pair is the one with smaller $\sigma_b\, Y$ (Y smaller for fewer teeth) ; design the pinion if $\sigma_{b1}\, Y_1 < \sigma_{b2}\, Y_2$, else the gear. Same material → pinion is weaker.
- σ_b allowable = $S_{ut}/3$ (static design stress, steel) ; endurance based (fatigue) lower ; typical : C45 normalised 140 MPa, hardened 200, FG 200 : 55.

9. Design for Dynamic Load & Wear Load

Dynamic (velocity) factor

$$F_t = P / v \quad ; \quad v = \pi\, d_1\, N_1 / 60 \ \text{(m/s)}$$

= 3 / (3 + v) (v < 7.5 m/s , ordinary cut) ; 6/(6 + v) (v up to 12.5 , carefully cut) ; 5.55/(5.55 + √

Effective load (Barth / Lewis) : $F_{eff} = C_s\, F_t / C_v \le S_b$

Service factor C_s : uniform load 1.0-1.25 ; light shock 1.25-1.5 ; moderate 1.5-1.75 ; heavy 1.75-2.25.

Buckingham dynamic load

$$F_d = F_t + F_i \quad , \quad F_i = 21\ v\ (b\ C + F_t) / (21\ v + \sqrt{(b\ C + F_t)})$$

Design : $S_b \geq F_d$ (with 1.25 - 1.5 factor of safety)

F_i = incremental load from tooth errors e (mm) and elasticity ; C = deformation factor = e / (1/E_1·...) listed for materials, e.g. error e = 0.05 mm steel-steel 20° → C ≈ 700 N/mm. v in m/s, b in mm. Higher accuracy gear → smaller e → smaller F_i.

Wear (surface) strength - Buckingham

$$S_w = b\ Q\ d_1\ K \quad \geq \quad F_d \quad (\text{or } F_{eff})$$

$Q = 2\ z_2 / (z_1 + z_2) = 2i/(i + 1)$ (external gears) ; $K = 0.16\ (BHN/100)^2$ N/mm² (20

- K = load-stress factor = $\sigma_{es}^2 \sin\varphi\ (1/E_1 + 1/E_2) / 1.4$ where σ_{es} = surface endurance limit = 2.75 BHN - 70 (N/mm²). Use the lower BHN of the pair. Q = $2z_2/(z_2 - z_1)$ for internal gears.

Spur gear design procedure

- (1) P, N_1, i. (2) $z_1 \approx 20$, $z_2 = i\ z_1$; select materials → σ_{b1}, σ_{b2}. (3) Y_1, Y_2 → weaker gear. (4) $b = 10\ m$; $v = \pi\ m\ z_1\ N_1/60000$; $F_t = P/v$; C_v. (5) $S_b = F_{eff}$ → solve m ; take standard module. (6) d_1, d_2, b, a. (7) Check dynamic load $F_d \leq S_b$ and wear $S_w \geq F_d$. (8) Check lubrication & hardness ; sketch with dimensions.

Example 3 : 15 kW, N_1 = 1000 rpm, i = 3, z_1 = 20 , z_2 = 60 , 20° full depth, σ_{b1} = 180 (pinion) , σ_{b2} = 120 MPa (gear), BHN 250, C_s = 1.25. Y_1 = 0.484 - 2.87/20 = 0.341 ; Y_2 = 0.436 → σY : pinion 61.4 ; gear 52.3 → gear weaker.

$b = 10\ m$; $v = \pi\ m \times 20 \times 1000/60000 = 1.047\ m$; $F_t = 14324/m$; $C_v = 3/(3 + 1.047\ m)$. $S_b = 120 \times 10m \times m \times 0.436 = 523\ m^2$. Equate : $523\ m^2 = 1.25 \times (14324/m)(3 + 1.047\ m)/3$ → $m^3 - 11.9\ m - 34.2 = 0$ → m = 4.4 → m = 5 mm (standard).

$d_1 = 100$, $d_2 = 300$, b = 50 mm , a = 200 mm. v = 5.24 m/s ; F_t = 2865 N ; C_v = 0.364 ; $F_{eff} = 1.25 \times 2865/0.364 = 9840$ N < $S_b = 120 \times 50 \times 5 \times 0.436 = 13090$ N (OK).

Wear : $Q = 2 \times 60/80 = 1.5$; $K = 0.16\ (2.5)^2 = 1.0$ → $S_w = 50 \times 1.5 \times 100 \times 1.0 = 7500$ N < 9840 (not OK) → raise hardness to BHN 350 : K = 1.96 → S_w = 14700 N (OK) (surface hardened).

Example 4 (trial $m = 4$) : same drive (15 kW, $N_1 = 1000$, $z_1 = 20$, $i = 3$), both gears hardened to BHN 400 → $K = 0.16 \times 4^2 = 2.56$ N/mm². $m = 4$: $d_1 = 80$, $b = 40$, $v = 4.19$ m/s , $F_t = 3581$ N , $C_v = 0.417$ → $F_{eff} = 1.25 \times 3581/0.417 = 10730$ N.

Wear : $S_w = b\,Q\,d_1\,K = 40 \times 1.5 \times 80 \times 2.56 = 12290$ N ≥ 10730 (OK). Bending (gear, $\sigma_b = 120$) : $S_b = 120 \times 40 \times 4 \times 0.436 = 8370$ N < 10730 (not OK) → bending governs, so take $m = 5$ as in Example 3.

- Lesson : check both beam strength and wear strength - hardness helps wear ($K \propto BHN^2$) but not bending (depends on σ_b and Y) ; the larger module required decides.

Typical exam questions - Unit V

- (1) Explain the modes of lubrication & Stribeck curve ; define bearing modulus. (2) Petroff's equation and McKee's equation ; heat balance. (3) Design a journal bearing for given W, N, d.
- (4) Stribeck's equation for static load ; L_{10} life & load-life relation ; select a ball bearing from catalogue. (5) Lewis equation, derivation, Buckingham dynamic and wear loads ; design a pair of spur gears.

Topic	Key formula
Petroff	$\mu = 2\pi^2 (ZN/p)(r/c)$
Heat balance	$\mu\, W\, v = C\, A\, (t_b - t_a)$
Stribeck	$C_0 = k\, d^2\, z\, /\, 5$
Bearing life	$L_{10} = (C/P)^3$ (ball)
Lewis	$W_t = \sigma_b\, b\, m\, Y$
Wear	$S_w = b\, Q\, d_1\, K$